

International AS and A-level

Biology (9610)

BL02 Biological systems and disease

Report on the examination

June 2024

REPORT ON EXAMINATION: INTERNATIONAL AS BIOLOGY (9610) BL02 JUNE 2024

Many students produced excellent responses to this paper whilst others found it challenging. There was no evidence that students were unable to complete the whole script within the time allowed.

The students found some command verbs easier to deal with than others. In general, the command verbs, 'give', 'name' and 'state' resulted in correct answers. Many students scored well in questions using the command verb 'describe' when it applied to a familiar context (02.1, 04.2, 06.1, 07.1). However when describe was used in the context of an unfamiliar application (03.3) students handled it less well. As expected, students found questions using explain (05.3) and evaluate more difficult. Interestingly, students found questions asking them to give evidence for and against more accessible than those asking them to evaluate, despite testing a similar skill.

Some students found it difficult to identify appropriate data to use in their answers (02.4, 06.4), whilst others ignored instructions to use data (02.3). Many students struggled with the difference between evidence (02.4) and conclusion (02.5, 04.5). However, most were able to suggest suitable controlled variables (04.4).

Maths questions were less well handled than previously. Incorrect reading of values from graphs (01.4, 04.3) cost marks. The ability to explain statistical tests (02.5) proved challenging. A number of students incorrectly referred to results being significant (02.5, 04.5).

A question on the required practical produced variable responses. Students were very clear as to why the root tip should be used (03.1) but less clear as to why toluidine blue is a suitable stain (03.2). Most were able to identify the stage when mitosis stops but were less able to give a reason for their choice (03.4).

BL02 ends with two questions requiring continuous prose. In general, students described the passage of water from the soil to the xylem very clearly (07.1). In question 07.2, the students were asked to relate the structure of the ileum to its function of absorption. Whilst there are signs of improvement in this skill over previous years, students still find it challenging. In this question students showed knowledge of the gross structure. However far fewer considered features at the cellular level.

QUESTION 01

This question considered the immune system and vaccine development. It included some straightforward starter questions and a simple calculation.

01.1

A straightforward opening question that was well-answered with most students correctly naming two suitable organ systems. The most common incorrect answer was to name an organ rather than the system.

01.2

This question on the process of phagocytosis was answered well with over 75% of students scoring 3 or 4 marks. Almost all students correctly gave the terms engulfed and lysosomes. If marks were lost, it was more likely students did not know receptors and/or lysozymes/hydrolytic enzymes.

01.3

Students were given a free choice to name three cells from the immune system together with their function. Therefore it is surprising that some students selected cell types that they then did not know the function of. Antibody production by B cells/plasma cells was the most common correct answer. Some students did not think to name both helper T cells and cytotoxic T cells, despite these cells having different functions. Many students that selected cytotoxic T cells lost the mark. This is because they incorrectly stated that the cytotoxic T cells killed the pathogen, rather than killing infected host cells.

01.4

This was expected to be a simple calculation requiring one value from the graph. Despite this, only around half of students scored. Errors included misreading from the graph or incorrectly copying of the value given in the text. Some students were unable to work with standard form.

01.5

This was a more challenging question designed to get the students to relate the topic to the real world. Many resorted to ideas of antigenic variation or that the vaccine itself would cause meningitis. Some students simply repeated that only 1% of cases develop meningitis. It should be stressed to students that simply repeating information given in the question will not score marks.

QUESTION 02

This question started by considering aphids and their feeding. The students were then presented with novel data about how plants might resist aphids.

02.1

Many students were able to describe how aphids feed. One omission was to not name the stylet, despite this being stated in the specification. A more troubling error was the suggestion that the aphids suck the sap from the plant.

02.2

Very few students failed to score this mark. The most common answer was sucrose, with far fewer naming amino acids. Some students gave sugar which is not specific enough at A-level.

02.3

The majority scored 1 or 2 marks on this question. Some students did not attempt any calculation and scored 1 mark. The statement 'Use data' and the large blank space above the answer lines should have alerted students to the need to measure the trichomes. Some students did not know how to use the scale bar. This skill should be practised.

02.4

In this question, students were presented with a table of data. Overall it was well-answered, with the majority scoring 1 or 2 marks. As expected, the best answers identified trends in the data. Common ones were that plants with G trichomes had fewer aphids but that plants with many total trichomes had more aphids. However, some students simply repeated the data. Others selected at random, two varieties to compare. In both cases the trend was not identified. A small number of students gave a conclusion rather than evidence.

02.5

This question asked the students to explain what can be concluded from a statistical test. More than half of the students failed to score either mark. Disappointingly, many students are still stating that data is significant. The test used was Spearman rank providing a correlation coefficient. When students scored 1 mark it was usually for identifying the negative correlation. The students found it more difficult to deal with the P value. Although they knew the significance of a P value below 0.05, they did not tailor their answer to the test used. Several gave the learned response that the difference was significant.

QUESTION 03

This question used the required practical on root tip squash to observe mitosis.

03.1

The majority scored on this straightforward one mark question.

03.2

This question on staining was generally well-answered. Common mistakes included stating the stain made the cell visible rather than the chromosomes. Another mistake was to give the reason why toluidine blue was a safer stain than acetic orcein. What the students said was often correct but clearly not the answer asked for in the question.

03.3

This question proved more challenging than expected. The students were asked about data collection and how a percentage value was calculated. Many simply assumed it was mitotic index. However, the Figure legend clearly stated that this was not the case. Others did identify which data to collect but forgot the need to multiply by 100 to gain a percentage.

03.4

Most students correctly identified a suitable stage of mitosis. However, they found it more difficult to explain why they had chosen that stage.

03.5

The students found it difficult to suggest what might be going wrong in the cell to stop mitosis at the stage they had identified in 03.4. Common incorrect suggestions included that enzymes are denatured by the cold. This suggests a fundamental misunderstanding of the effect of low temperature on enzyme activity.

03.6

Next, the students were asked why a certain time for incubation was chosen in the second experiment. Many answers were vague, suggesting that this was the 'best' time. Very few considered possible negative effects of keeping the beans at low temperature for many hours.

03.7

Students were asked to evaluate the decision to combine data in the second experiment. Many students focused on minor differences in the three stages that were combined, rather than looking at the shared trend. Many seemed to think that combining would make it 'easier' to collect or analyse the data. Few looked at the graph for the second experiment.

03.8

This was not well answered. The students were directed to use two different figures from the two different experiments. However, many were unclear as to which experiment they were talking about. Few students viewed the decrease in the percentage of cells in prophase as an indicator of the resumption of mitosis.

QUESTION 04

This question was based on another of the required practicals, the effect of exercise on heart rate.

04.1

Students generally understood why non-smokers were chosen for the experiment. They also understood the idea of a resting heart rate. However, few went on to the idea of a baseline measurement before exercise.

04.2

There were some very good answers to this question. However, there was also a range of scores between 0 and 4 marks. Some poor answers simply copied values from the graph. The graph was slightly unusual with negative values on the y-axis. The majority understood that this meant blood was leaving the ventricle. However, some ignored the negative values and stated that blood was entering the ventricle. Most students knew that the ventricle was contracting. Most also knew that the blood would leave via the aorta. Fewer mentioned any of the valves. Some students gave lengthy descriptions involving the SAN and bundle of His. This information is not on the specification until BL04 so will never appear on a BL02 exam paper.

04.3

Surprisingly, students found calculating cardiac output difficult. Most knew the equation so were able to score 1 mark. However, fewer scored all 3 marks. The most common mistake was to assume 1 heartbeat took 1000 milliseconds.

04.4

Most students found this question simple and were able to score both marks. Students favoured more generic answers like age and gender. Fewer students gave answers specifically related to the investigation.

04.5

Students tended to make comparisons between the three sports only. Fewer considered comparing with the starting values. Some students simply repeated the results shown in the graph, rather than concluding if an increase was significant or not. Disappointingly, the phrase, the results are significant was seen far too often.

04.6

A spread of scores was seen on this question, but few students scored all 3 marks. Some wasted time describing the graph, before starting their answers. Many students simply stated that the mass of the left ventricle had increased, without linking this to more muscle and therefore more powerful contractions. Stroke volume was rarely seen. The idea of more oxygen was commonly seen, but some missed out on this marking point by not indicating where the oxygen was needed or for what purpose.

QUESTION 05

This question used the familiar context of the digestive system to present some novel data for analysis.

05.1

The vast majority of students correctly answered this multiple choice question.

05.2

Students favoured the idea of physical digestion over mixing with enzymes. However, quite a number thought that physical digestion occurs at the molecular level rather than simply breaking large pieces of food into smaller pieces.

05.3

This question proved very challenging. Very few students were able to use the information and graphs to arrive at the idea that higher levels of gastrin would cause the secretion of more pepsin. Many focused only on gastrin or reported that gastrin and pepsin were higher in infected than not infected patients. Very few students were able to use the information to suggest a mechanism by which the high levels of the protease enzyme could cause the ulcers.

05.4

This question resulted in many generic answers about correlation is not causation or the role of unknown factors. Despite a large number of possible marking points, by far the most common correct answer was the idea of the overlap of the gastrin levels between infected and not infected patients.

QUESTION 06

This question considered how oral rehydration therapy works, before moving on to questions about experimental design.

06.1

This question produced lots of excellent 3-mark answers. All marking points were seen, approximately equally. Very few students failed to mention osmosis, although there was occasionally confusion about the direction of water movement. Similarly, as some students failed to differentiate between the small intestine, epithelial cells and the lumen, it was unclear where the sodium ions were being absorbed.

06.2

Students often find this kind of evaluate question more approachable. Many cited the idea that dilutions reduced the concentration of sodium ions as a reason not to dilute the ORS. Far fewer were able to suggest that the ORS might still work as effectively as a reason against. Several students stated that the more ORS in the dilution, the more sodium ions there were. This was used in both the for and against arguments. Again this suggests that students have not really grasped the difference between results and conclusions.

06.3

Students found it quite difficult to provide 2 specific answers to this question. The most common correct answer was the idea of taste being subjective. Some students spotted that the subjects were

a mix of adults and children or that they did not have the condition. Since both these facts are given in the question, students were expected to add more in order to score. Generic answers such as small sample size (not the case) or no repeats were often given. Students should be discouraged from doing this in favour of answers specific to the question.

06.4

The majority of students suggested that as the dilutions tasted worse it is not a good idea to dilute the ORS. Far fewer considered the likely effectiveness of the dilutions by comparing their sodium ion concentrations with the minimum values given by the WHO recommendations. Those that did use the WHO values tended to refer to 75 mmol dm^{-3} for fast recovery. A disappointing number of students referred to sodium rather than sodium ions (or Na^+) or quoted values of 75 or 60 without any units. Students should be discouraged from both these errors as they risk being penalised.

QUESTION 07

The final question consisted of two extended answers. One about the absorption of water from the soil into a plant and one about absorption in the ileum.

07.1

The question was answered well with over 60% scoring 3 marks or more. On the whole, students presented a logical progression of how water moves from the soil to the xylem. The vast majority discussed osmosis. Most then went on to name the apoplastic and symplastic pathways (although a few did mix up which pathway went through the cell wall/cytoplasm). Many students knew about the Casparian strip but did not know its position at the endodermis. Students were able to explain about active transport of minerals lowering the water potential of the xylem. Despite the statement not to, some students wasted time describing the movement of water up the xylem.

07.2

In general, students found relating structure to function more challenging. As expected, they scored less well on this question than on 07.1. All marking points were seen although few mentioned the long length of the ileum. Most commonly students started with the large surface area provided by the villi and microvilli. Those that did not score this marking point usually mentioned one or the other, but not both. Next most popular was the idea of a good blood supply. Far fewer students considered features at the cellular level such as carrier proteins or many mitochondria. Most errors were seen with the idea of a reduced diffusion pathway. Students referring to the ileum itself rather than the walls of the villi/capillaries being thin.